

Table 3: Metrics for linear models and principal components over a dataset of numbers of varying integer places with mixed signs.

Model	Provider	Linear R^2		PCA R^2		PCA Variance	
		Min	Max	Min	Max	Min	Max
gemini-embedding-001	Google	0.48 ± 0.08	0.99 ± 0.01	-0.03 ± 0.02	0.72 ± 0.02	0.02 ± 0.00	0.10 ± 0.00
text-embedding-3-large	OpenAI	0.91 ± 0.02	1.00 ± 0.00	0.69 ± 0.03	0.84 ± 0.02	0.10 ± 0.00	0.19 ± 0.00
text-embedding-3-small	OpenAI	0.89 ± 0.01	1.00 ± 0.00	0.41 ± 0.07	0.82 ± 0.01	0.07 ± 0.00	0.20 ± 0.00
text-embedding-ada-002	OpenAI	0.90 ± 0.01	1.00 ± 0.00	0.59 ± 0.03	0.79 ± 0.01	0.09 ± 0.00	0.17 ± 0.00
voyage-3-large	Voyage	0.95 ± 0.02	1.00 ± 0.01	0.73 ± 0.01	0.83 ± 0.02	0.28 ± 0.00	0.44 ± 0.00
voyage-3.5	Voyage	0.93 ± 0.01	1.00 ± 0.00	0.67 ± 0.02	0.85 ± 0.02	0.29 ± 0.00	0.38 ± 0.00
voyage-3.5-lite	Voyage	0.90 ± 0.06	1.00 ± 0.00	0.68 ± 0.02	0.87 ± 0.02	0.26 ± 0.00	0.42 ± 0.00

ues, comes to represent only the sign of the numbers (see Figure 6). The introduction of negative values consistently degrades performance metrics, indicating that models struggle with the concept of negative numbers more than might be expected. This limitation suggests caution when using embeddings for datasets containing both positive and negative values.

Third, embedding models introduce systematic noise into numerical representations, with those based on large language models being particularly prone to this issue. The low explained variance ratios demonstrate that much of the embedding space is devoted to capturing information orthogonal to the numerical content itself. This noise may interfere with applications requiring precise numerical relationships, such as mathematical reasoning or quantitative analysis tasks.

Conclusion

We have conducted a comprehensive evaluation of numerical precision in LLM embedding models, examining how well these systems encode scalar values across different ranges and precisions. Our analysis reveals that while embedding models can preserve numerical information sufficiently for linear reconstruction, they introduce substantial noise that limits their effectiveness for precise numerical applications.

The key finding is that embedding models exhibit a fundamental trade-off between preserving numerical information and introducing extraneous variation. While the primary component of variation in embeddings does correlate with numerical values, the majority of the embedding space encodes information unrelated to the numerical content. This suggests that current embedding models, despite their success in many natural language processing tasks, may not be optimal for applications requiring precise numerical understanding.

Future work should focus on developing embedding architectures specifically designed for numerical data, potentially through specialized pretraining objectives or architectural modifications that better isolate numerical information from other sources of variation. Additionally, investigating techniques for denoising numerical embeddings or identifying the most relevant dimensions for numerical tasks could improve the practical utility of existing models for quantitative applications.

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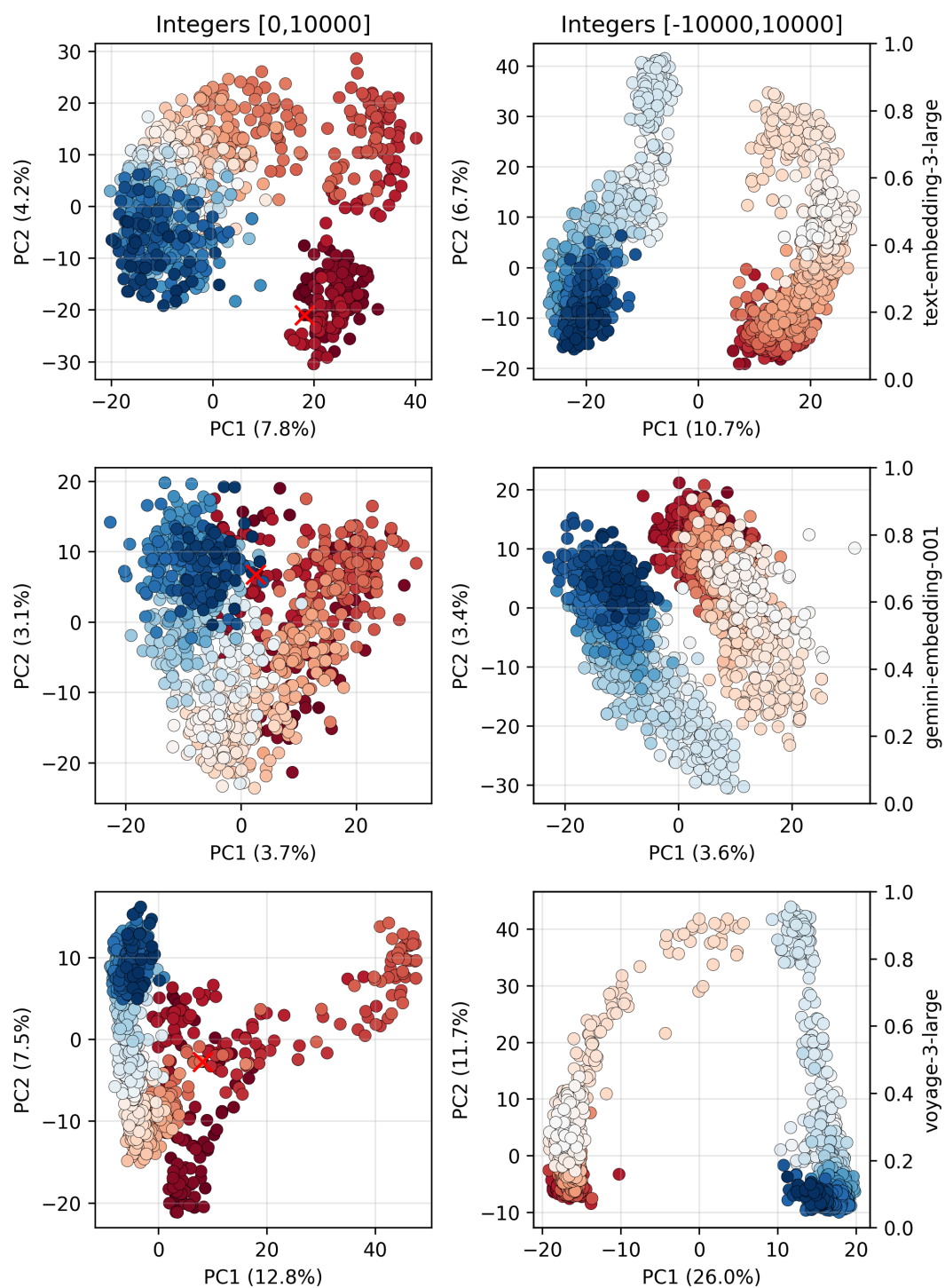


Figure 7: PCA over randomly sampled $x \in [0, 10k]$, $|X| = 1000$ (left) and $x \in [-10k, 10k]$, $|X| = 2000$ (right) for the main OpenAI, Voyage and Gemini models.